

Example 7: Find a basis for the column space of the matrix

$$B = \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution: Denote the columns of B by $b_1, \dots, b_5$.

Observation: $b_3 = -3b_1 + 2b_2$ and $b_4 = 5b_1 - b_2$.

The fact that b_3 and b_4 are combinations of the pivot columns means that any combination of the ~~pivot~~ columns $b_1, \dots, b_5$ is actually just a combination of b_1, b_2 and b_5 .

Indeed, if v is any vector in $\text{Col } B$, say, $v = c_1b_1 + c_2b_2 + c_3b_3 + c_4b_4 + c_5b_5$ where $c_1, c_2, c_3, c_4, c_5 \in \mathbb{R}$, then by substituting b_3 and b_4 in v , we have

$$v = c_1b_1 + c_2b_2 + c_3(-3b_1 + 2b_2) + c_4(5b_1 - b_2) + c_5b_5$$

$$\Rightarrow v = c_1b_1 + c_2b_2 + -3c_3b_1 + 2c_3b_2 + 5c_4b_1 - c_4b_2 + c_5b_5$$

$$= b_1(c_1 - 3c_3 + 5c_4) + b_2(c_2 - c_4 + 2c_3) + c_5b_5 \quad (\text{Skipping steps})$$

$\therefore v$ is a linear combination of b_1, b_2 and b_5 . So $\{b_1, b_2, b_5\}$ spans $\text{Col } B$.

Also, b_1, b_2 and b_5 are linearly independent, because they are columns from an identity matrix.

Important observation
 $\therefore$ The pivot columns of B form a basis for $\text{Col } B$. $\square$

The matrix B in Example 7 is in reduced echelon form. To handle a general matrix A , recall that linear dependence relations among the columns of A can be expressed in the form $Ax = 0$ for some x . (If some columns are not involved in a particular dependence relation, then the corresponding entries in x are zero.)

When A is row reduced to echelon form B , the columns are drastically changed, but the equations $Ax = 0$ and $Bx = 0$ have the same set of solutions. That is, the columns of A have exactly the same linear dependence relationships as the columns of B .